Midnight, Not Midday: Air Conditioning, Electric Scooter Charging and Distribution Transformer Ageing in a Lahore Neighbourhood

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***Abstract*— Pakistan plans to put about two million electric two-wheelers on the road by 2030, and most will be charged at home, on distribution transformers that already carry heavy air-conditioning load in summer. The studies of charging and transformer ageing reviewed here deal with cars in the United States, Europe and Australia, and the common advice to charge "at night, off peak" assumes that household load falls at night. This study tests that assumption with one year of measured one-minute household data from Lahore (PRECON). Every house uses more power as the day gets hotter (median 77 W per kelvin of daily mean temperature above about 22.5 °C, tested on unseen weeks), and a virtual feeder of 34 houses peaks around midnight from April to September, not in the afternoon. Scooter charging from two use profiles is added under five charging rules, and the transformer's hot-spot temperature and insulation ageing are computed with the IEC 60076-7 thermal model. With a scooter in every house (4 % more energy), charging on arrival raises yearly ageing by 21 %; a timer that starts charging when the tariff peak window ends raises it by 36 %, because it lands on the night peak; a timer that finishes charging just before the morning departure raises it by 8 %, the same as an ideal valley-filling schedule. On a tighter transformer the same rules give +50 %, +110 % and +11 %. A climate 2 K warmer, applied through the measured temperature slope, raises baseline ageing 2.4 times, more than any charging rule.**

***Index Terms*—** *distribution transformer, electric two-wheeler, smart charging, air conditioning, PRECON, IEC 60076-7, Pakistan*

I. Introduction

Pakistan's National Electric Vehicle Policy 2025–30 counts about 2.05 million two-wheelers in its target of 2.21 million new energy vehicles [1]. E-scooters sold locally have small packs and chargers (the Eveon Pronto, for example, has a 72 V, 32 Ah pack and a 4 A charger [2]), so they are charged from ordinary house sockets. Their load therefore lands on the same 11/0.4 kV distribution transformers that serve residential air conditioning (AC), which drives the summer load in the data used here (Section VI-A).

The insulation of an oil-immersed transformer ages with its hot-spot temperature, which depends on the load, the ambient temperature and the timing of both [3]. A charging rule that moves load in time can therefore change ageing even when it adds no energy. This has been studied in detail for electric cars (Section II), and the usual conclusion is that uncontrolled evening charging raises ageing and that moving it later helps. Two things make the Pakistani case different: the vehicles are small two-wheelers with sub-kilowatt chargers, and summer nights in Lahore are hot, so household AC may keep the night load high.

This paper asks two questions. When is a Lahore neighbourhood transformer most loaded, according to measured household data? And for the same scooter energy, how much does the time of charging change the transformer's yearly ageing? It uses PRECON [4], one year of one-minute power data from houses in Lahore, with NASA POWER weather [5], and extends two earlier studies by the author: a home PV and battery schedule for Karachi [6] and a thermal-cycling study of scooter motor controllers [7], whose scooter model supplies the charging energy here.

Section II reviews related work and Section III states the gap and the contributions. Sections IV and V give the data and the models, Section VI the results, and Section VII the limitations.

II. Related Work

A. EV charging and distribution transformer ageing

Gong *et al.* [8] combined a dynamic thermal model of a residential distribution transformer with Monte Carlo charging scenarios to estimate how plug-in vehicle charging affects transformer life. Hilshey *et al.* [9] used the IEEE C57.91 model for a 25 kVA transformer serving 12 homes and compared a cool climate (Burlington, Vermont) with a hot one (Phoenix, Arizona). Uncontrolled Level 2 charging raised ageing by a factor of 9.79 in Phoenix against 2.34 in Burlington, so the ambient temperature mattered as much as the charging itself. They also found that simply delaying Level 2 charging until after midnight increased ageing, and that a smart charging scheme that used the transformer temperature reduced ageing by a factor of 4.8 to 6.3. Soleimani and Kezunovic [10] used the same IEEE model with Nissan LEAF and Bolt charging at 40 % penetration and proposed a fuzzy-logic smart charging scheme to limit loss of life. Clement-Nyns *et al.* [11] looked at a Belgian residen-

tial grid with Belgian residential load profiles; uncoordinated evening charging gave the worst voltage deviations, and at 30 % penetration their test feeder needed a larger transformer.

Rooftop PV interacts with these results. Geiles and Islam [12] found that rooftop PV can offset part of the ageing caused by vehicle charging. Behi *et al.* [13] applied the IEC thermal model to a 200 kVA transformer on a Perth feeder; PV lowered loss of life by about 58 % by relieving off-peak loading, but did not reduce the peak, because the peak and PV output did not coincide, and PV with demand response together cut loss of life by about 90 %.

B. Timing of home charging

At a larger scale, Muratori [14] showed that uncoordinated home charging reshapes residential demand and that clustering of vehicles raises local peaks. Powell *et al.* [15] found that more daytime charging reduces grid impacts, and that home charging scheduled locally to off-peak hours can itself create new peaks. Li and Jenn [16] modelled California's distribution networks and found that about half of the feeders would need upgrades by 2035 under uncontrolled charging, with residential feeders needing about twice the upgrades of commercial ones and residential overloads occurring at night. A recurring message across these studies is that a timer that sends every vehicle to the same "off-peak" hour can move the problem rather than solve it.

C. Temperature and household demand

Auffhammer *et al.* [17] showed for the United States that peak electricity demand is more sensitive to temperature than average demand, so warming matters most at the peak. Change-point regression, in which load is flat below a balance temperature and rises linearly above it, is a standard way to measure such sensitivity from metered data [18].

For Pakistan, PRECON [4] is the main open source of high-resolution household data. Its authors later used ambient temperature as an input for short-term load forecasting on the same data, where regression and support vector models did better than a neural network, and noted load shedding in the records [19]. Bajwa *et al.* [20] used PRECON within a nine-city climate study and found that AC takes a mean 21.4 % of household electricity, rising above 33 % in June and July; their household analysis reports monthly shares and does not model load against temperature, time of day, or the network.

D. Electric two-wheelers

Abdul-Manan *et al.* [21] found that in India electric two-wheelers already reduce life-cycle greenhouse gas emissions compared with petrol ones and recommended prioritising them, noting that their small, removable batteries can be charged indoors or swapped. Corti *et al.* [22] reviewed charging infrastructure for electric micromobility and noted its lower power and the lack of a standard output voltage. Huu and Ngoc [23] proposed a load-profile tracking algorithm for two-wheeler charging stations with PV in Vietnam, aimed at filling valleys and cutting peaks. Table 1 summarises the studies closest to this one.

Table 1. Studies closest to this work

Study	Vehicles	What it shows
Gong [8]	cars	dynamic thermal model with Monte Carlo charging
Hilshey [9]	cars	hot climate ageing ≈4× cool; after-midnight delay can raise ageing
Soleimani [10]	cars	fuzzy smart charging to limit loss of life
Behi [13]	none (PV, DR)	IEC model; PV helps off-peak, not the peak
Li, Jenn [16]	cars	half of feeders need upgrades by 2035; residential overloads at night
Huu, Ngoc [23]	two-wheelers	station scheduling with PV; no transformer ageing
Bajwa [20]	none	PRECON AC share by month; no network
This work	two-wheelers	measured 1-min Lahore load, IEC ageing, Pakistani tariff timer

III. Research Gap and Contributions

From the studies above, three gaps stand out.

1. Transformer-ageing studies of home charging that were found deal with cars with charging power of several kilowatts, in the United States, Europe and Australia. The two-wheeler studies found deal with emissions, charger design or charging stations, not with the residential transformer.

2. The advice to charge at night assumes a night valley in household load. Hilshey *et al.* [9] and Li and Jenn [16] already show that night charging can overload US transformers, but no study found here checks the shape of the night load in a hot South Asian city with measured data. Work on PRECON so far forecasts load or reports AC shares [19, 20]; it does not ask when a shared transformer is most loaded.

3. No study found evaluates a timer tied to Pakistan's time-of-use peak windows [25] against transformer ageing.

This paper makes four contributions. (i) It measures the temperature sensitivity of 41 PRECON houses and of a 34-house virtual feeder, with a test on held-out weeks. (ii) It shows that this feeder peaks near midnight in summer and that the summer tariff peak window ends just as the feeder's own peak begins. (iii) It compares five charging rules, including the UK default random delay and a NEPRA-window timer, by IEC 60076-7 insulation ageing, and shows that the rule matters more than the energy. (iv) The code, derived data and an interactive tool are open, so every number can be reproduced.

IV. Data

A. Household data

PRECON gives one-minute kW for each house from June 2018 to May 2019, with a total-load column and, in 33 houses, one or more metered AC circuits [4]. House 35 was not in the downloaded archive, so 41 houses are used. The files contain no

missing minutes, although the dataset paper states that the meters switch off during power cuts [4]; the gaps were therefore filled in some way the paper does not describe. Stretches of 10 minutes or more in which all columns of a house lie on one exact straight line were treated as filled and removed from every fit. They are 0.4–29 % of a house's minutes (median 1.6 %). Whether they are power cuts, logger outages or both cannot be told from the data, so no outage analysis is attempted.

B. Weather

Hourly 2 m air temperature and global horizontal irradiance for Lahore (31.52 N, 74.36 E) come from NASA POWER [5], in local solar time. POWER defines local solar time in 15° bands, so at this longitude it equals Pakistan Standard Time. Each hourly value is the mean over the hour that begins at its time stamp, so it is placed at mid-hour before linear interpolation to minutes. The file has no missing values; the hottest hour was 47.5 °C (May 2019).

V. Methods

A. Temperature sensitivity

For each house, daily mean load P is fitted against daily mean temperature T with the change-point model [18]

$$P = a + b\max(0,\ T - T_b), \tag{1}$$

with T_b searched on a 0.5 K grid and a, b found by least squares. Days need at least 20 valid hours. The model is fitted on even ISO weeks and tested on odd weeks; the reference for the test R^2 is the house's mean load.

B. Virtual feeder

The 41 houses are spread over Lahore and are not on one transformer. As the study's main assumption, houses are summed minute by minute as if they were. Filled minutes are replaced by the same house's mean for that month and clock hour, which models the feeder as if there were no load shedding. Seven houses with any month less than half measured are left out of the feeder (they stay in the temperature fits), leaving 34 houses and 1.7 % imputed energy. The feeder load is converted to apparent power with an assumed power factor of 0.95.

C. Scooter charging

The scooter is the 102 kg, 72 V model of [7], driving part 1 of the WMTC cycle with the road load of UN GTR No. 2 and an assumed motor and controller efficiency of 0.88. It needs 21.3 Wh/km at the battery; as a rough check, the Pronto's pack and claimed 90 km range give about 26 Wh/km [2]. With an assumed charger efficiency of 0.85, a commuter (24 km/day, Monday to Saturday) needs 0.60 kWh/day from the grid and a delivery rider (96 km/day, every day) 2.41 kWh/day. The base case has one scooter per house, 20 % of them delivery riders, and 0.5 kW chargers. The Pronto's 4 A charger at a 72 V pack is about 0.3 kW at the pack, or about 0.35 kW from the grid, so 0.35 kW and 1 kW are tested as well. Commuters plug in between 18:00 and 19:30 and leave at 07:45; delivery riders plug in between 22:30 and 23:30 and leave at 10:45 (uniform random times). Every scooter must get back that day's energy before it leaves. Five charging rules are compared:

- *plug-in*: charge at full power on arrival;
- *random delay*: as plug-in, after a random delay of up to 600 s, the default in the UK's smart charge point rules [24];
- *after peak*: start when the time-of-use peak window ends. NEPRA's LESCO tariff sets the windows as 17:00–21:00 (December–February), 18:00–22:00 (March–May and September–November) and 19:00–23:00 (June–August) [25];
- *latest*: start so that charging ends exactly at departure;
- *valley fill*: charge in the lowest-load minutes before departure, knowing the feeder load in advance. This minimises load, not ageing, and is an ideal reference rather than a practical rule.

D. Transformer thermal model

The top-oil temperature θ_o and hot-spot temperature θ_h follow the difference equations of IEC 60076-7, Annex C [3], with load factor K (apparent power over rating) and ambient θ_a:

$$D\theta_o = \frac{Dt}{k_{11}\tau_o}\left[\left(\frac{1+K^2R}{1+R}\right)^{x}\Delta\theta_{or} - (\theta_o - \theta_a)\right], \tag{2}$$

$$D\Delta\theta_{h1} = \frac{Dt}{k_{22}\tau_w}\left[k_{21}\Delta\theta_{hr}K^{y} - \Delta\theta_{h1}\right], \tag{3}$$

$$D\Delta\theta_{h2} = \frac{Dt}{\tau_o/k_{22}}\left[(k_{21}-1)\Delta\theta_{hr}K^{y} - \Delta\theta_{h2}\right], \tag{4}$$

$$\theta_h = \theta_o + \Delta\theta_{h1} - \Delta\theta_{h2}. \tag{5}$$

The values are those given for ONAN distribution transformers [3,26]: $x = 0.8$, $y = 1.6$, $k_{11} = 1$, $k_{21} = 1$, $k_{22} = 2$, $\tau_o = 180$ min, $\tau_w = 4$ min, $R = 5$, $\Delta\theta_{or} = 55$ K and $\Delta\theta_{hr} = 23$ K. The step Dt is 1 min, below half the smallest time constant as the standard requires, and the first two days are run twice as a warm-up. Air temperature is the ambient. Ageing uses the relative ageing rate for non-upgraded paper,

$$V = 2^{(\theta_h - 98)/6}, \tag{6}$$

summed to equivalent hours at 98 °C per year [3]. The base rating is 100 kVA, with 75 and 200 kVA as alternatives; how many such houses a real 100 kVA unit serves is not known here, so results are given relative to a no-scooter run on the same rating.

E. Scenarios

Study A runs the feeder without scooters on the three ratings. Study B raises the share of houses with a scooter from 0 to 100 % in steps of 25 %. Study C repeats the full-adoption case for other ratings, charger powers, 1.5 times the energy, only commuters, half delivery riders, rooftop PV on 30 % of houses (3 kWp each from horizontal irradiance with a performance ratio of 0.8, an assumption) and another random seed. Study D

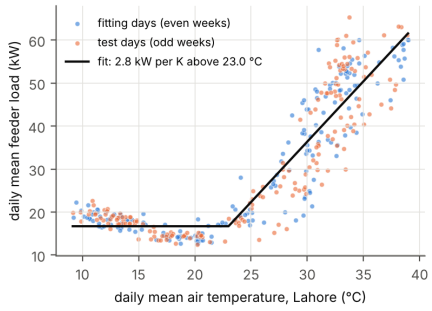


Figure 1. Daily mean feeder load against daily mean air temperature, 34 houses, June 2018 to May 2019. The change-point fit uses even weeks only.

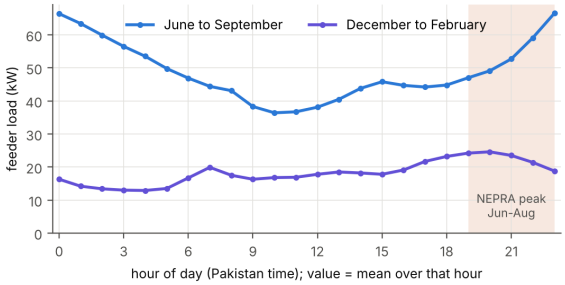


Figure 2. Mean feeder load by hour of day in summer and winter. Shaded: the June–August time-of-use peak window [25].

adds 2 K to the air temperature and, on days above the balance temperature, adds load through the measured feeder slope, spread over the day in proportion to that day's load.

VI. Results

A. Heat and household load

Across 41 houses the median slope is 77 W/K (interquartile range 44–120 W/K) above a median balance temperature of 22.5 °C. On the unseen test weeks the median R^2 is 0.68 and the median absolute error halves, from 0.39 kW without temperature to 0.19 kW. The metered AC circuits alone give a median 38 W/K (R^2 0.45), about half of the total, because not every AC unit in a house is on a metered circuit. For the 34-house feeder the slope is 2.8 kW/K above 23 °C, with a test R^2 of 0.84 (Fig. 1).

B. When the feeder is most loaded

The feeder uses 266 MWh a year. Its highest one-minute load, 101 kW, came at 23:50 on 4 June 2018, and from April to September the hour with the highest mean load is 23:00 or midnight (Fig. 2). From June to September the mean load is 66 kW at midnight and falls to a minimum of 36 kW at 10:00, while the afternoon, the hottest part of the day, sits in between. In winter the peak moves to 19:00–20:00 and the minimum to 04:00. The summer tariff peak window (19:00–23:00) therefore ends just as the feeder's own peak begins. This is the household version of the night overloads that Li and Jenn [16] found for car charging, but here it is present before any vehicle is added.

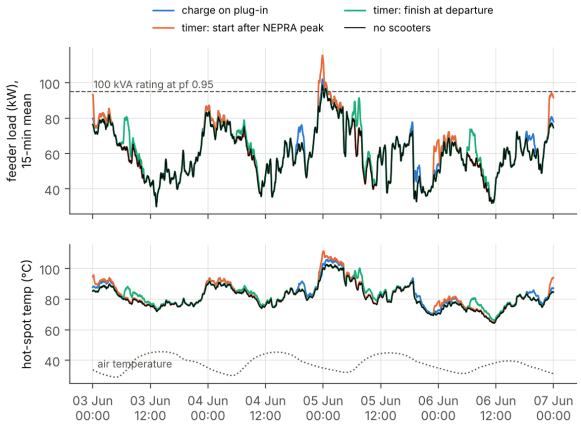


Figure 3. Feeder load (top) and hot-spot temperature (bottom), 3–6 June 2018, 100 kVA, a scooter in every house.

Table 2. Yearly ageing with a scooter in every house, relative to no scooters on the same transformer

Rule	100 kVA	75 kVA	peak (p.u.), 100 kVA
no scooters	1.00	1.00	1.06
plug-in	1.21	1.50	1.10
random delay	1.21	–	1.10
after peak	1.36	2.10	1.24
latest	1.08	1.11	1.06
valley fill	1.08	1.11	1.06

C. Baseline transformer ageing

On 100 kVA the baseline peak is 1.06 per unit, the load exceeds rating for 1.7 h in the year, and the hot spot reaches 102.6 °C, below the 120 °C limit for normal cyclic loading [3]. Yearly ageing is 200 equivalent hours, 90 % of it from June to September. On 75 kVA, which is the same as about 45 such houses on 100 kVA, ageing is 2528 h and the hot spot exceeds 120 °C on 16 days (23.2 h). On 200 kVA it is 26 h. The rating assumption thus changes absolute ageing by two orders of magnitude, which is why the results below are relative.

D. Charging rules

A scooter in every house adds 11.3 MWh a year, 4 % of the feeder's energy. Fig. 3 shows the effect during the hottest days, and Fig. 4 and Table 2 the yearly result.

Charging on arrival adds 21 %, mostly from delivery riders who plug in at 22:30–23:30, on top of the night peak. The 600 s random delay changes almost nothing: it spreads starts over minutes, while the transformer responds over hours. The after-peak timer is the worst rule: it moves commuters from the early evening to 23:00 and stacks every scooter at one time, so ageing rises 36 % and the time above rating goes from 1.7 to 26 h. This is the same effect that Hilshey *et al.* [9] found for an after-midnight delay of car charging. The "latest" rule puts charging in the early morning, when the load is falling and the air is coolest, and gives the same yearly ageing as the ideal

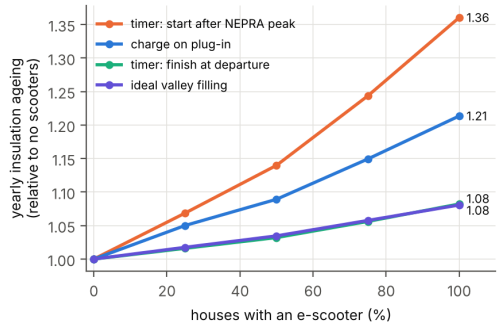


Figure 4. Yearly insulation ageing against the share of houses with a scooter, 100 kVA.

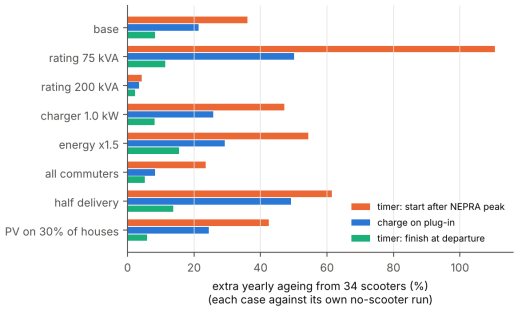


Figure 5. Extra yearly ageing from a scooter in every house under other assumptions, each against its own no-scooter run.

valley fill. On 75 kVA the order is the same and the gaps are larger: the after-peak timer doubles ageing and raises the time above 120 °C from 23.2 to 85.0 h, against 26.7 h for the latest rule.

E. Sensitivity

Fig. 5 repeats the comparison for other assumptions. The order after peak > plug-in > latest holds in every case. Delivery riders matter most: with half of the scooters used for delivery, plug-in charging adds 49 %. With only commuters, plug-in adds 8 % but the after-peak timer still adds 24 %. With 1 kW chargers the latest rule keeps ageing low but creates a new morning peak of 1.11 p.u. (07:28 on 5 June), because the delivery riders' scooters, which share one departure time, start together; real departure times would spread this. In a few cases the latest rule ages the transformer slightly less than valley fill (for example 216.9 against 217.2 h with 0.35 kW chargers), because valley fill minimises load and does not see the ambient temperature. Rooftop PV on 30 % of houses lowers baseline ageing by 20 % by cutting the hot afternoon load, but it does not touch the night peak, so the charging results change little (+24 %, +42 % and +6 %). This agrees with Behi *et al.* [13], where PV relieved off-peak loading but not the peak.

F. A warmer climate

Adding 2 K to the air temperature, and to the load through the measured feeder slope, adds 29.7 MWh (11 %) and raises baseline ageing from 200 to 487 h, 2.4 times. This is larger than the effect of any charging rule. With scooters added in the warmer case, plug-in charging gives 609 h, the after-peak timer 705 h and the latest rule 526 h. The warming case extrapolates a one-year relation and should be read as an order of magnitude, in line with the finding of Hilshey *et al.* [9] that climate can matter as much as the vehicles.

VII. Discussion and Limitations

The night peak comes straight from the measured data and does not depend on the transformer model. It changes the practical advice: in such a neighbourhood "charge at night after the peak" is the worst of the simple rules tested, and "finish just before you leave" is as good as the ideal schedule. A rule like that needs only the departure time and the energy to be charged, which a scooter charger could ask for.

The limits are real.

- The 34 houses are not one feeder and are mostly large houses with several AC units; a real street would differ in size, diversity and mix. Power factor, the number of houses per transformer and the charger and use profiles are assumptions, and only the scooter part is a model.
- Gap-filled minutes were imputed, and load shedding, which the dataset's authors noted in the records [19] and which in reality removes load in blocks, is not modelled.
- The train and test weeks alternate, so they share seasons and neighbouring days are correlated; the test R^2 is therefore an optimistic measure of how the fits would do in another year.
- Weather is hourly and was interpolated to minutes on the assumption that each value is the mean of the hour after its time stamp. Air temperature is used as the transformer's ambient, without sun on the tank.
- The temperature slope comes from one year and partly reflects seasonal habits, so the warming case is an extrapolation.
- The peak windows are from the 2026 tariff, not from 2018–19, and residential time-of-use applies only to connections with a peak load above 5 kW [25].

VIII. Conclusion

In one year of measured data from Lahore, household load rises by about 77 W per kelvin above 22.5 °C, and a 34-house feeder peaks near midnight in summer. On such a feeder, the time of scooter charging decides its effect on the transformer more than the energy itself: from +8 % to +36 % yearly ageing for the same 4 % more energy. Charging that ends at departure is as good as the ideal schedule, and charging that starts after the tariff peak is the worst of the rules tested. A climate 2 K warmer would matter more than any of them. Natural next steps are measured data from one real feeder, including load shedding, and a charging rule that also uses the transformer or air temperature, as proposed for cars in [9].

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